

# Semester — Master Implementation Brief

**Purpose:** Hand this document to Claude Code, ChatGPT, an engineering team, or an AI coding agent to extend the existing live Semester product safely and systematically.

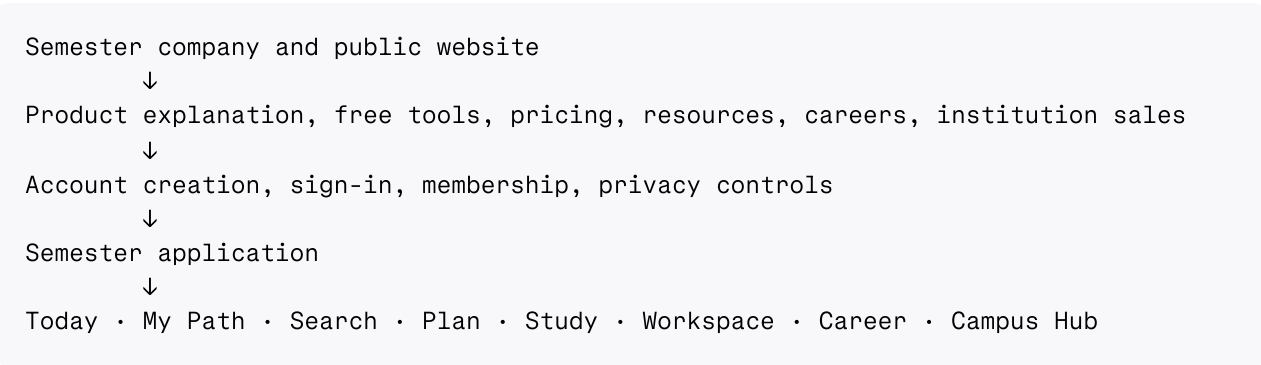
**Repository:** <https://github.com/harrisonjrubin7-cmyk/semester>

**Current live product:** <https://harrisonjrubin7-cmyk.github.io/semester/>

**Important:** This is an integration-and-extension project, **not** a ground-up rewrite. Preserve all working features, routes, design language, tests, and data behavior unless a migration is explicitly planned, tested, documented, and reversible.

## 1. Product mandate

Semester is both the company and the product. Build it as one unified ecosystem:



## Core promise

**College is complicated. Your path shouldn't be.**

Semester helps students understand their academic path, decide what to do next, plan registration, study course material, create work, access campus support, prepare for meetings, develop career evidence, and connect current choices to future goals.

## Positioning

**Semester is the academic navigation and momentum platform.**

Semester should not initially claim to replace every university system. It is the connected experience and action layer across systems:

| System                 | Authoritative purpose                                      | Semester purpose                                         |
|------------------------|------------------------------------------------------------|----------------------------------------------------------|
| Banner / Workday / SIS | Official records, catalog, enrollment, registration, holds | Explain, plan, route actions, show approved summary data |

| System                              | Authoritative purpose                                      | Semester purpose                                            |
|-------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|
| Canvas / Blackboard / Brightspace   | Course delivery, submissions, grades, instructor workflows | Connect approved context/materials, study tools, deep links |
| Top Hat / Gradescope / Turnitin     | Engagement, grading, assessment/originality workflows      | Surface permitted context and official deep links           |
| Housing / dining / student accounts | Contracts, balances, official transactions                 | Show verified status, deadlines, and safe handoffs          |
| Google / Microsoft                  | Files, email, calendars, collaboration                     | Connect selected student-authorized materials and events    |
| Zoom / Teams / Calendly             | Meetings and recordings                                    | Prepare agendas, meetings, follow-up tasks                  |
| Semester                            | Navigation, planning, study, creation, actions, goals      | Daily student home and cross-system coordination layer      |

## 2. Current repository reality

The repository already has substantial foundations. Do not duplicate them before auditing.

### Known repository structure

```

semester/
├── app/
│   ├── src/
│   │   ├── App.tsx
│   │   ├── main.tsx
│   │   ├── screens.tsx
│   │   ├── screens/
│   │   ├── components/
│   │   ├── state/
│   │   ├── data/
│   │   ├── ai/
│   │   ├── intelligence/
│   │   ├── insights/
│   │   ├── ally/
│   │   ├── lib/
│   │   └── styles/
│   ├── api/
│   ├── server/
│   ├── scripts/
│   ├── public/
│   ├── package.json
│   ├── vite.config.ts
│   ├── vercel.json
│   ├── .env.example
│   └── README.md
├── supabase/
├── docs/
├── packages/
├── project/
└── pipeline/

```

```
|— .claude/
|— .github/
|— ACTION-PLAN.md
|— APP-AUDIT.md
|— ENGINEERING-AUDIT.md
|— FEATURE-INVENTORY.md
|— IMPLEMENTATION-PLAN.md
|— PILOT.md
|— SECURITY.md
|— SECRETS.md
|— MONITORING.md
|— STAGING.md
|— ROLLBACK.md
|— REGRESSION-CHECKLIST.md
|— TOPHAT.md
|— GRADESCOPE-TURNITIN.md
|— SEMESTER_IMPLEMENTATION_PLAN.md
|— SEMESTER_IMPLEMENTATION_STATUS.md
|— SEMESTER_PRODUCT_COMPLETENESS_MATRIX.md
|— README.md
```

The app appears to be a Vite + TypeScript application with extensive state/data modules, tests, Supabase scaffolding, accessibility code, AI/intelligence modules, and Vercel configuration. The existing state model is substantial, including `app/src/state/shape.ts`, `store.tsx`, reducers, persistence modules, and test coverage. Existing data modules include campus, catalog, events, fellowships, onboarding, institutional preview, seed data, schools, courses, and transcript-related sources.

## First instruction to the implementing agent

Before changing code, conduct a **read-only repository audit** and create `docs/AUDIT.md`. Inspect:

1. `app/package.json` and build scripts.
2. `app/README.md`, `README.md`, `.env.example`, `vercel.json`, `vite.config.ts`.
3. Existing route/screen system in `App.tsx`, `main.tsx`, `screens.tsx`, and `screens/`.
4. Navigation registry and shortcuts.
5. Existing state store, reducers, shape definitions, persistence, and `localStorage` usage.
6. Supabase folders, migrations, client configuration, Auth status, RLS status, Storage status, and Edge Functions.
7. Existing AI and Intelligence modules.
8. Accessibility utilities and tests.
9. Existing data mocks, fixtures, seed data, and whether sample data is clearly labeled.
10. Existing tests, CI, deployment, Vercel setup, branches, and open pull requests.
11. Existing implementation-status, security, pilot, and feature documents.
12. Any secrets accidentally in tracked files. Do not print secrets in audit output.

The audit must state:

- Current framework and build command.
- Current deploy target and branch.
- Current routes/screens.
- Existing completed product capabilities.
- Existing mock/demo behavior versus production-backed behavior.
- LocalStorage/mock-to-Supabase migration plan.
- Auth and authorization status.
- Biggest security risks.
- Exact recommended file-level work for Sprint 1.
- Open questions requiring human decision.

### 3. Non-negotiable implementation rules

1. **Do not rewrite the application wholesale.** Extend and refactor in small vertical slices.
2. **Do not delete working routes or code** without a migration plan, tests, rollback note, and explicit approval.
3. **Never expose secrets in browser code.** All AI keys, Stripe keys, integration credentials, and service secrets remain server-side.
4. **Do not claim official data is verified** unless it came from an approved current source.  
Display source labels:
  - Institution verified
  - Imported
  - Student entered
  - Estimated
  - Needs review
5. **Use progressive disclosure.** Do not bury students in every capability at once.
6. **Keep top-level student navigation to five destinations:** Today, My Path, Search, Plan, Me.
7. **Every recommendation must be explainable:** why it appeared, source data used, expected impact, limitations, alternatives, and controls.
8. **All consequential writes require explicit confirmation:** send, share, export, schedule, create payment handoff, delete, register, add/drop, or modify a record.
9. **Use feature flags** by tenant, role, pilot cohort, and release stage.
10. **Build WCAG 2.2 AA toward the baseline.** Keyboard access, visible focus, screen-reader labels, text resizing, high contrast, captions/transcripts, reduced motion, accessible math/figures, and no drag-only essential controls.
11. **Build source-aware data models** with source system, record ID, last sync, confidence, access classification, owner, and retention policy.

12. Integrate first; replace only when appropriate. Do not recreate SIS, LMS, healthcare, payment, or official registration authority without institutional agreements and maturity.

## 4. Product information architecture

### Primary student destinations

| Today       | My Path                           | Search                                                                                           | Plan | Me |
|-------------|-----------------------------------|--------------------------------------------------------------------------------------------------|------|----|
| Destination | Student job                       | Contents                                                                                         |      |    |
| Today       | Understand what matters now       | Path Snapshot, Action Center, Next Best Step, urgent commitments, study prompt                   |      |    |
| My Path     | Plan degree and registration      | Requirements, degree map, term plan, scenarios, registration readiness, study abroad impact      |      |    |
| Search      | Ask, discover, and act            | Courses, requirements, people, resources, jobs, events, departments, policies, opportunities     |      |    |
| Plan        | Organize time and commitments     | Schedule, course drafts, study blocks, calendar, tasks, meetings, travel                         |      |    |
| Me          | Manage personal work and controls | Profile, Study Library, Workspace, Career, Portfolio, membership, privacy, settings, connections |      |    |

### Contextual modules

Do not create permanent top-level destinations for every niche need. Open modules when context requires them:

- Study Studio / Course Companion.
- Semester Workspace.
- Math & Data Lab.
- Writing Studio.
- Career Hub.
- Campus Hub.
- Housing / Dining / Student Accounts cards.
- Athlete Plan.
- Study Abroad.
- Supporter View.
- Faculty Course Studio.
- Advisor Meeting Mode.

## 5. The canonical Action Center

Build a shared Action Center across every module. It is the primary coordination layer for Semester.

### Action model

```
type ActionStatus =
  | 'open'
  | 'in_progress'
  | 'blocked'
  | 'snoozed'
  | 'completed'
  | 'dismissed'
  | 'expired'
  | 'cancelled'

type ActionPriority = 'critical' | 'high' | 'normal' | 'low'

type ActionSourceKind =
  | 'institution_verified'
  | 'sis'
  | 'lms'
  | 'faculty'
  | 'advisor'
  | 'student'
  | 'integration'
  | 'ai_generated'
  | 'estimated'

interface SemesterAction {
  id: string
  tenantId: string
  studentId: string
  type: string
  title: string
  summary: string
  whyItMatters: string
  status: ActionStatus
  priority: ActionPriority
  dueAt: string | null
  recommendedStartAt: string | null
  expiresAt: string | null
  estimatedMinutes: number | null
  source: {
    kind: ActionSourceKind
    system: string
    recordId: string | null
    sourceUrl: string | null
    label: string
    verifiedAt: string | null
    syncedAt: string | null
    confidence: 'verified' | 'high' | 'medium' | 'low' | 'needs_review'
  }
  explanation: {
```

```

    trigger: string
    factors: string[]
    expectedImpact: string | null
    limitations: string[]
    sourceReferences: Array<{ label: string; url?: string; recordId?: string }>
  }
  primaryAction: {
    label: string
    kind: 'internal_route' | 'external_deep_link' | 'draft' | 'review' | 'request' |
    target: string
    requiresConfirmation: boolean
    requiresFreshAuth: boolean
    requiredPermissions: string[]
  }
  secondaryActions: Array<{ label: string; kind: string; target?: string }>
  links: Record<string, string | undefined>
  privacy: {
    visibility: 'private' | 'shared_with_advisor' | 'shared_with_supporter' | 'insti
    containsSensitiveData: boolean
    dataClassification: 'public' | 'internal' | 'student_private' | 'restricted'
  }
  delivery: {
    inApp: boolean
    pushAllowed: boolean
    emailAllowed: boolean
    notificationState: 'not_sent' | 'sent' | 'seen' | 'suppressed'
    fatigueScore: number
  }
  lifecycle: {
    createdAt: string
    updatedAt: string
    completedAt: string | null
    completedBy: 'student' | 'institution_sync' | 'advisor' | 'system' | null
    dismissedReason: string | null
    snoozedUntil: string | null
    version: number
  }
}

```

## Supporting persistence

```

actions
action_sources
action_explanations
action_links
action_delivery
action_interactions
action_status_history
action_permissions
action_audit_events
action_feedback
action_rules

```

## Action Center UX rules

- Show one **Most Important** action.
- Show only three to five Next actions before **View all**.
- Group related work: “Registration readiness: two steps remaining.”
- Rank: urgency + goal/academic impact + actionability + source confidence – notification fatigue.
- Use three layers: Notice → Understand → Act.
- Show source and freshness.
- Let student snooze, dismiss, correct, share, or ask for help.
- Never describe students as “at risk,” “failing,” or “behind.”
- Update actions from verified external events without overwriting personal student notes or plans.

## 6. Sprint 1: Registration and Path Pilot

### Goal

Build the smallest real product that proves Semester’s core promise:

A student can create an account, see a credible Path Snapshot, build a next-term plan, resolve basic schedule conflicts, save backup courses, prepare for an advising conversation, and know what to do next.

### Target users

Pick one initial cohort:

- First-year students before registration.
- Transfer students.
- Advising-center students.
- Student athletes.
- One department/program with complex prerequisites.

### Include

- Secure account creation and sign-in.
- Email verification and account recovery.
- Basic student profile: school, program, target graduation term, completed credits, goals.
- Today dashboard.
- Action Center.

- Path Snapshot.
- Student-entered or curated program/requirement data clearly labeled as estimate unless verified.
- Basic term planner.
- Course list/search for pilot cohort.
- Weekly schedule and conflict detection.
- Registration readiness checklist.
- Backup course shortlist.
- Advisor meeting agenda and view-only share/export.
- Feedback prompt: “Did this help you understand what to do next?”
- Support / report issue route.
- Source labels and data freshness.
- Mobile responsive and accessible states.
- Analytics for activation and meaningful task completion.

## Explicitly defer

- Production Banner/Workday write integrations.
- Official registration execution.
- Payment/billing checkout.
- Housing/dining/health integrations.
- Emergency Disclosure Console.
- Full parent/supporter experience.
- Broad social network.
- Full Career/Campus Hub.
- All Study Studio formats.
- Unrestricted AI agents.
- Deep gradebook integrations.
- Complex institution analytics.

## First success metrics

Account creation rate  
 Onboarding completion rate  
 First Path Snapshot rate  
 Term-plan creation rate  
 Backup-course saved rate  
 Schedule-conflict resolution rate  
 Advisor-agenda creation rate  
 First meaningful action completion rate

"I understand what I need to do next" survey score  
Advisor usefulness feedback  
Support issue / failed-search themes

## 7. Today dashboard

### Mobile layout

Good afternoon, Maya  
Friday, September 25

#### YOUR PATH

On track for May 2028 based on your current plan  
84 / 120 credits · 70% complete  
2 choices need review  
[View My Path]

#### NEXT BEST STEP

Choose a quantitative-reasoning course for Fall.  
Because: one requirement remains open in your plan.  
[Explore options] [Why am I seeing this?]

#### UPCOMING

Today – Statistics lecture, 11:00 AM  
Today – Prepare advisor questions, 4:00 PM  
In 3 days – Chemistry quiz  
In 9 days – Registration opens

#### QUICK ACTIONS

Search · Build plan · Add course · Schedule · Prepare for advising

### Desktop layout

Left rail: Today · My Path · Search · Plan · Me

#### Main workspace:

- Greeting
- Path Overview
- Next Best Step
- Plan Health
- Career/Growth card
- Recently viewed/saved

#### Right context pane:

- Upcoming calendar
- Deadlines
- Selected-item impact
- Data source/freshness
- Scenario state
- Advisor context

## Recommendation explanation sheet

Why this recommendation?

We suggested adding a quantitative-reasoning course because:

- Your plan has one open requirement.
- Your Fall plan has room for three credits.
- Adding an eligible course supports your May 2028 target.

Data used:

- Your saved plan
- Planned Fall courses
- Your target graduation term

Limitations:

- Course availability and official eligibility must be confirmed by your institution

[View requirement] [Show alternatives] [Not relevant right now]

## 8. My Path, registration, and planning

### Registration Center flow

```
Official SIS/catalog data or curated pilot data
  ↓
Student enters goals and constraints
  ↓
Semester identifies eligible pathways
  ↓
Student compares courses and builds schedule
  ↓
Semester checks planning constraints
  ↓
Student saves primary and backup plan
  ↓
Student optionally shares advisor agenda
  ↓
Student deep-links to official registration system
```

### Checks

| Check               | Authority                    | Semester behavior                                            |
|---------------------|------------------------------|--------------------------------------------------------------|
| Registration window | SIS                          | Display verified time when connected                         |
| Hold category       | SIS                          | Display high-level category and official action route        |
| Prerequisite        | Catalog/SIS                  | Display source and planning interpretation                   |
| Seat availability   | Official registration system | Show timestamped data only if approved; never guarantee seat |
| Degree requirement  | Official audit/catalog       | Label verified only when connected                           |

| Check                 | Authority                   | Semester behavior                           |
|-----------------------|-----------------------------|---------------------------------------------|
| Schedule conflict     | Semester plan               | Calculate and explain                       |
| Credit load           | Semester plan               | Calculate planned credits                   |
| Workload/time balance | Student-entered constraints | Optional guidance, not official restriction |

## Side-by-side desktop views

- Requirement detail + eligible course options.
- Weekly schedule + selected course impact.
- Current plan + proposed scenario.
- Shared plan + advisor meeting agenda.

Limit comparisons to three options. Show top three relevant options, then Show all eligible options.

## 9. Study Studio and Course Companion

### Core promise

Upload or connect a syllabus, notes, slides, documents, PDFs, audio, video, readings, and assignment instructions. Semester organizes approved materials, identifies what matters, creates source-linked study assets, and lets the student choose how to study.

### Material pipeline

```

Select/upload/connect source
↓
File validation + malware scanning + format detection
↓
Text/OCR/table/figure/transcript extraction
↓
Preserve page, slide, timestamp, source file version, and access rights
↓
Extract headings, dates, objectives, assignments, key terms, citations
↓
Student/faculty review extraction
↓
Index authorized source material
↓
Generate source-grounded study assets

```

## **Syllabus extraction**

Extract for review, never auto-apply without student confirmation:

- Course name/section.
- Instructor and office hours.
- Class meetings.
- Grade breakdown.
- Assignments, exams, projects, and deadlines.
- Required materials.
- Attendance, late-work, and make-up policies.
- Accessibility/support statements.
- Learning outcomes.
- Weekly timeline.

## **Fourteen study modes**

1. Comprehensive Guide.
2. Quick Summary.
3. Bullet Notes.
4. Flashcards.
5. Practice Quiz.
6. Exam Simulator.
7. Study Plan.
8. Case Study.
9. Worked Examples.
10. Concept Map.
11. Visual Explainer.
12. Audio Study Guide.
13. Video Recap/Lesson.
14. Teach-Back Mode.

Do not claim students have fixed learning styles. Offer flexible modalities based on subject, goal, time, accessibility needs, explicit preference, and low-stakes practice feedback.

## Self-quizzing

Every question must show answer/rubric, explanation, source anchors, and user feedback controls.

Question 4 of 10

Research Methods • Week 4

Which conclusion is supported by a positive correlation?

[Answer choices]

Correct: C

Why: Correlation identifies a relationship but does not establish causation.

Sources: Week 4 slides, slide 12; course review guide, section 2.

[Open source] [Try similar] [Add to review] [I guessed] [I do not understand]

## Academic integrity

- Respect faculty/course AI rules.
- Do not answer prohibited active graded assessments.
- Offer explanation, analogous practice, study planning, and source review instead.
- Never fabricate citations, research, data, quotes, lab results, or sources.
- Generated writing/slides are drafts and require student review.

## 10. Workspace, Docs, Slides, Data, and Meetings

### Semester Workspace

My Files

Course Libraries

Notes

Docs

Slides

Tables

Study Studio

Calendar

Meetings

Email Drafts

Shared With Me

Connected Accounts

## Docs

Create editable, source-aware drafts for:

- Study guides.
- Essay outlines.
- Research plans.
- Lab reports.
- Literature reviews.
- Case studies.
- Project proposals.
- Advisor agendas.
- Resume bullets.
- Professional emails.
- Scholarship planners.

## Slides

- Import PDF/PPTX/selected Google Slides.
- Extract headings, notes, figures, charts, tables, and citations.
- Build editable outlines and drafts.
- Include speaker notes, citations, alt-text prompts, contrast checks, rehearsal timer, transcript/captions.
- Export to Google Slides, PowerPoint, PDF, or outline.

## Tables / Math & Data Lab

- Grade calculator, assignment tracker, lab-data organizer, budget, application tracker.
- CSV/Excel/Sheets import with student-selected data.
- Descriptive statistics, distributions, correlations, regression, confidence intervals, hypothesis-test visualizations, accessible charts.
- Require source dataset and expose assumptions/limitations.
- Never fabricate data, p-values, participants, results, charts, or research conclusions.

## Meetings

- Create agendas, selected plan links, questions, materials, and follow-up actions.
- Integrate selected Google/Outlook/Calendly/Zoom meetings through authorized connections.
- Do not automatically send messages, record meetings, import recordings, or summarize sensitive discussions without explicit authorization.

## 11. STEM, Writing, and Athlete support

### STEM

- Equation-aware notes.
- Worked solutions with reasoning and common mistakes.
- Adaptive practice problems.
- Formula/reference sheet builder.
- Function plots, scatterplots, histograms, box plots, regression, residuals, probability/distribution visualizers, confidence interval and hypothesis-test visualizers.
- Accessible charts: labels, units, alt text, source, export, method notes.
- Lab Companion: pre-lab checklist, approved safety sources, data capture, calculation, report structure, figure captions.
- Coding support: repository links, explanation, debugging, tests, architecture diagrams, but no prohibited complete graded-project delivery.

### Writing Studio

- Prompt/rubric parser.
- Research question and source plan.
- Evidence/quote/paraphrase organizer.
- Thesis and argument map.
- Outline, paragraph plan, counterargument builder.
- Citation support and “citation needed” prompts.
- Revision checklist tied to rubric.
- Instructor feedback interpreter.
- Read aloud, accessibility checks, presentation conversion.
- No hidden ghostwriting, plagiarism evasion, fabricated sources, or prohibited assessment completion.

### Athlete Plan

- Overlay practice, travel, competition, training, work, courses, deadlines, and study blocks.
- Travel-aware early-start recommendations.
- Downloadable offline study packs.
- Faculty communication drafts for student review.
- Tutoring around travel window.
- Student-controlled share with approved academic-support staff.

- Never expose health/injury/recovery data or use app activity as an eligibility/effort score.

## 12. Career, jobs, internships, applications, and portfolio

### Career Hub

Career explorer  
Course-to-career map  
Skills inventory  
Portfolio  
Resume builder  
Cover letter workspace  
Internship and job tracker  
Career fair planner  
Mock interview studio  
Networking notes  
Mentor discovery  
Graduate-school planner  
Offer comparison

### Course-to-career map

Course outcomes  
↓  
Projects, assignments, labs, presentations, research  
↓  
Demonstrated skills  
↓  
Portfolio evidence  
↓  
Internships, jobs, graduate paths, career opportunities

Students must accept/edit/reject suggested skill mappings. Do not guarantee a job or invent skills.

### Resume templates

- Classic student resume.
- Internship resume.
- Research CV.
- STEM/technical resume.
- Creative portfolio resume.
- Leadership resume.
- Federal/public service resume.
- Graduate-school CV.

- Academic CV.
- Career-change resume.
- International configurable resume.
- One-page career fair resume.

## Application tracker

Saved → In progress → Interviewing → Completed

Track deadlines, required documents, selected resume version, cover letter, interview date, follow-up, and student-entered status. Do not apply automatically or invent claims.

## Mock interviews

- Behavioral/STAR practice.
- Role-specific and technical prompts.
- Case framework practice.
- Organization research checklist.
- Student-controlled voice/video practice.
- Pacing/filler-word feedback only if opted in.
- Thank-you draft after interview.
- No facial/emotion analysis, hiring prediction, or peer ranking.

## 13. Campus Hub

### Campus Hub categories

Housing & Residence Life  
 Dining & Meal Plans  
 Tuition & Student Accounts  
 Financial Aid & Scholarships  
 Campus Services  
 Department Sites  
 Events & Clubs  
 Study Abroad  
 Jobs & Internships  
 Transportation & Parking  
 Safety & Support  
 Important Dates

Semester displays an action layer; official services retain authoritative transaction functions.

## **Housing**

Show approved summary data, deadlines, checklists, contacts, move-in/move-out tasks, and official portal links. Do not show roommate data, full location details, accommodations, contract/payment detail, or sensitive records without authority.

## **Dining**

Show plan name, optional balance summary, plan-change deadline, locations, hours, menus, dietary labels, official wait-time estimate with source/freshness, and official mobile ordering deep link. Do not store item-by-item dining history or payment data.

## **Billing and financial aid**

Show deadlines, high-level account action/hold category, official portal link, payment-plan deadline, scholarship/aid checklist, and neutral reminders. Do not process payments initially. Do not expose detailed balances, account ledgers, bank/payment data, or aid determinations in generic Semester UI.

## **Department and university content**

Build a Campus Graph for departments, programs, faculty, labs, advising, support offices, events, policies, resources, and opportunities. All content needs owner, source, last-reviewed date, and verified state.

## **Clubs and events**

Ingest only approved campus/department/organization feeds via API, iCal, RSS/JSON, or managed CMS. Normalize event fields, support search/filters, calendar save, RSVP/deep-link, accessibility information, student interest controls, and no public popularity rankings.

## **14. Study abroad**

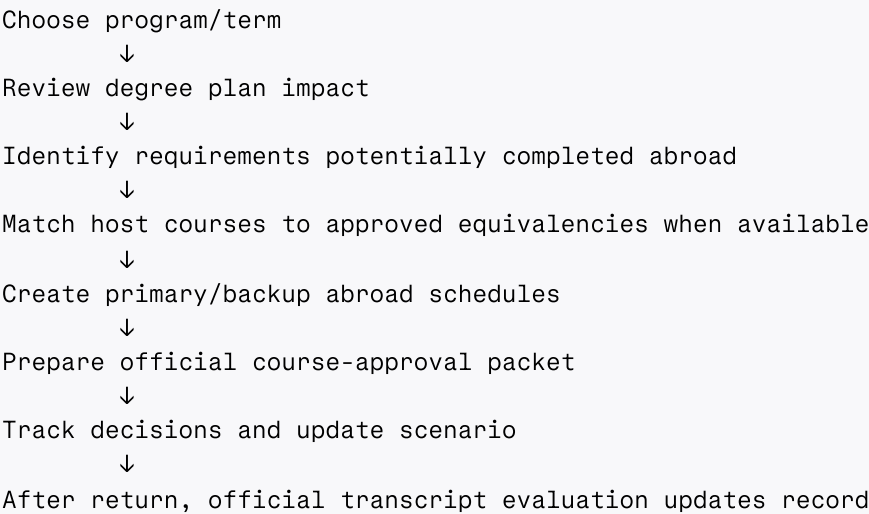
Study Abroad belongs inside My Path scenario planning.

## **Rule**

Never say a course will transfer without official approval. Labels must be:

- Pre-approved.
- Pending review.
- Estimated equivalent.
- Not evaluated.

# Workflow



Track program application, academic approval, financial-aid discussion, passport/visa, housing, insurance, pre-departure orientation, return transcript, and career reflection.

## 15. Roles and permissions

| Role                 | Purpose                                                          | Boundary                                                         |
|----------------------|------------------------------------------------------------------|------------------------------------------------------------------|
| Student              | Plan, study, search, create, connect accounts, selectively share | Owns personal plan/drafts/study activity/sharing choices         |
| Prospective student  | Explore majors/careers/readiness                                 | No official record assumed                                       |
| Faculty              | Materials, course guidance, AI policy, Course Studio             | No duplicate LMS/gradebook; no private plans by default          |
| Advisor              | Shared plan, agenda, authorized follow-up                        | No health/billing/private study/supporter information by default |
| Tutor/support        | Student-selected assistance                                      | Only assigned/consented context                                  |
| Parent/supporter     | Student-consented milestone view                                 | No default education-record access                               |
| Registrar/curriculum | Publish official catalog/requirements                            | Source-of-truth management                                       |
| Institution admin    | Tenant/configuration/aggregate data                              | No unrestricted record browsing                                  |
| IT/security          | Integrations, policies, audit                                    | Least-privilege audited access                                   |
| Semester support     | Support and technical troubleshooting                            | No student data absent documented authorization                  |

Use RBAC + ABAC + object-level authorization. Recheck authorization for each sensitive request.

## 16. Integrations

### Connection order

1. Semester account/authentication.
2. Google/Microsoft optional calendar/file connections.
3. Canvas LTI 1.3 launch and approved course context.
4. Institution SSO.
5. Read-only SIS/curriculum data.
6. Advising, event, resource, and calendar feeds.
7. Housing/dining/billing status/deep links.
8. Deeper enterprise integrations only after pilot maturity.

### Google and Microsoft

- Use OAuth 2.0 delegated scopes and incremental consent.
- Google Drive: use Picker and `drive.file` where possible; avoid broad Drive access.
- Microsoft: use delegated Graph permissions and request the smallest permission set.
- Calendar: read access first; request write only at the moment a student chooses to create/update an event.
- Gmail/Outlook: draft messages for student review; never automatically send.
- Display connected account, granted scopes, selected files, last sync, disconnect, and delete-imported-copies controls.

### Canvas/LMS

- Use LTI 1.3 / LTI Advantage.
- Start with secure launch, identity/course context, deep links back to official Canvas work.
- Import only approved metadata/files/materials.
- Do not ingest grades, attendance, activity, or submissions by default.
- Gradescope/Turnitin/Top Hat remain authoritative for their functions; Semester surfaces approved student-facing context and deep links.

### Banner/Workday

- Use institution-owned integration layer, API gateway, Ellucian Ethos, Workday APIs, RaaS, or approved middleware.
- Do not connect directly to production SIS databases.
- Minimal initial fields: active status, program, term, enrollment, catalog/course data, registration window, hold category, advisor assignment if authorized.
- Label source/freshness and provide official fallbacks.

## System-to-system security

```
Semester service ↔ institution API gateway  
mTLS + OAuth client credentials or signed JWT + short-lived token + scopes + tenant :
```

Use webhooks with signature verification, timestamp checks, replay protection, durable inbox, idempotency key, version ordering, audit history, and scheduled reconciliation.

## 17. Security, privacy, reliability, and emergency boundaries

### Zero trust requirements

- SAML/OIDC for users.
- MFA for admin and high-risk actions.
- OAuth 2.0 delegated scopes for connected student accounts.
- mTLS + short-lived workload identity for sensitive service-to-service APIs.
- Least privilege and object-level authorization.
- Tenant isolation.
- Encryption in transit and at rest.
- Secrets manager, rotation, no secrets in code.
- Signed webhooks, schema validation, WAF/rate limits, replay prevention.
- Audit logs for sensitive access/write/share/export actions.
- Backups, restore tests, monitoring, error alerting, incident response, staging/production separation.

### Privacy commitments

- Do not sell student data.
- Do not use student data for advertising.
- Do not train public models on student data without explicit documented authorization.
- Support export, deletion, disconnect, source visibility, permission revocation, and sharing controls.
- Do not treat clicks, time in app, LMS activity, or study time as proof of ability, motivation, wellbeing, or risk.

### Emergency Disclosure Console

Do not build this in the student MVP. It is enterprise-only, institution-governed, separately authenticated, field-limited, just-in-time, fully audited, and only for legally authorized emergency responders. It must not appear in Supporter View or normal admin tools.

# 18. Semester Intelligence

## Four modes

| Mode     | Purpose                                                                |
|----------|------------------------------------------------------------------------|
| Analyze  | Extract, summarize, identify patterns in selected authorized materials |
| Explain  | Teach source-grounded concepts in requested format                     |
| Generate | Build editable study assets, drafts, plans, slides, notes              |
| Execute  | Prepare safe product actions after explicit preview and confirmation   |

## Model architecture



Never hard-code the product to a single provider. Select provider/model based on capabilities, approved institutional policy, data sensitivity, cost, latency, and user settings.

## Required AI controls

- Source citations/page/slide/timestamp anchors.
- Source access checks.
- Confidence/missing-source warnings.
- Model/provider policy transparency where required.
- Course/assignment AI rules.
- Human confirmation for writes/sends/shares/exports/schedules.
- Output history/delete controls.
- Prompt-injection defenses for uploaded content.
- AI feedback/report issue path.
- No official degree certification or registration decisions by AI.

## 19. Public website, company, careers, and membership

### Public route set

```
/
/product
/students
/institutions
/solutions
/pricing
/tools
/resources
/stories
/about
/careers
/contact
/security
/privacy
/accessibility
/help
/status
/changelog
/login
/signup
/account
/membership
/app/*
```

### Homepage promise

**Headline:** College is complicated. Your path shouldn't be.

**Copy:** Semester brings degree requirements, course options, schedule, goals, and next steps into one intelligent home—so students can make every semester count.

**Primary CTA:** Start planning free.

**Secondary CTA:** See how it works.

**Institution CTA:** Explore Semester for institutions.

### Careers

**Headline:** Help make college easier to navigate.

Include opportunities for front-end, full-stack, back-end, mobile, product design, accessibility, academic advising, student success, campus ambassadors, content creators, partnerships, and advisors.

**Initial contact email:**

`harrisonjrubin7@gmail.com`

Future aliases:

```
hello@semester.com
support@semester.com
careers@semester.com
partners@semester.com
security@semester.com
press@semester.com
```

Membership

| Plan            | Suggested price            | Package                                                                             |
|-----------------|----------------------------|-------------------------------------------------------------------------------------|
| Free            | \$0                        | Profile, one plan, basic schedule, directory, saved goals/reminders                 |
| Plus            | \$7.99/month or \$59/year  | Unlimited plans, comparisons, scenarios, calendar sync, sharing/export              |
| Pro             | \$14.99/month or \$99/year | What-if scenarios, major/minor comparison, portfolio, advanced career/advisor tools |
| Semester Access | Institution sponsored      | SSO, verified data, campus directory, configured premium features                   |

Users must be able to upgrade, downgrade, cancel, export data, request deletion, manage payment method, view invoices, and check institutional eligibility. Never hide cancellation or paywall personal data export/deletion.

20. Institution deployment and pilot model

First institutional offer

Semester Registration and Path Pilot

```
Length: 2-6 weeks
Cohort: 25-100 students
Use case: registration readiness and academic planning

Semester provides:
• student onboarding
• Path Snapshot
• term/course planning
• schedule/conflict view
• registration action checklist
• advisor agenda
• student clarity survey
• advisor feedback collection
• outcomes report

Institution provides:
• pilot sponsor
• cohort invitation
```

- basic program/course/deadline information
- staff champion
- feedback participation

## Institutional rollout stages

1. Discover: goals, cohort, stakeholders, data scope, risks, success metrics.
2. Configure: branding, roles, content, SSO, minimal approved data.
3. Validate: data quality, security, accessibility, workflows, support.
4. Launch: train staff/students and run campaign.
5. Measure: adoption, planning, clarity, advisor usefulness, support outcomes.
6. Expand: add cohorts, modules, and integrations after evidence.

## 21. Metrics and research

### North-star metric

**Weekly Path Progressed Students:** a student who completes at least one meaningful planning, studying, support, career, or advising action in a week.

### Measure separately

| Layer              | Examples                                                                                        |
|--------------------|-------------------------------------------------------------------------------------------------|
| Engagement         | Logins, searches, page opens, time in app                                                       |
| Meaningful use     | Plan saved, backup course, conflict resolved, agenda created, quiz completed                    |
| Student experience | Clarity, confidence, agency, support discovery, trust                                           |
| Operations         | Support tickets, resolution time, self-service success                                          |
| Outcomes           | Registration timing, credit pace, persistence, completion—only with approved careful evaluation |

### Core student question

“I understand what I need to do next.”

Do not claim Semester causes retention, grades, persistence, or graduation without rigorous privacy-governed research design.

## 22. Feature governance

Create a Feature Governance Board: product, design, engineering, accessibility, privacy/data, security, student research, institutional success, and relevant domain owner.

Every feature requires:

Feature name  
Target user  
Job to be done  
Trigger/moment  
Current workaround  
Smallest solution  
Data required  
Authoritative source  
Permissions  
Accessibility acceptance criteria  
Privacy/security risk  
Success metric  
Operational owner  
Failure/fallback behavior  
What it replaces or merges  
Out of scope  
Sunset condition

#### Rules:

- Five primary student destinations only.
- One Action Center, one Search, one Workspace, one Intelligence layer.
- Contextual modules appear only when relevant.
- Quarterly feature review: invest, simplify, merge/hide, defer, retire.
- No named owner, metric, support plan, or accessibility plan = do not launch.

## 23. Implementation sequence

### Phase 0: Foundation

- Repository audit.
- Production data model.
- Authentication/persistence.
- Tenant model.
- Privacy controls.
- Monitoring/errors/backups.
- Responsive and accessibility baseline.
- Feature flags.
- Staging/production separation.

## **Phase 1: Student daily value**

- Today + Action Center.
- My Path + Registration readiness.
- Search + Ask + Act.
- Basic Plan/Schedule.
- Calendar connection/export.
- Advisor Meeting Mode.
- Basic Study Studio.
- Public web, signup, membership, help/feedback.

## **Phase 2: Educator/advisor value**

- Faculty Course Studio.
- Canvas LTI launch.
- Faculty-approved study packs.
- Course AI rules.
- Advisor shared planning.
- Campus directory/content governance.
- Career Hub.

## **Phase 3: Institution confidence**

- SSO, SCIM, institution administration.
- Minimal Banner/Workday feed.
- Aggregate pilot analytics.
- Security/trust/procurement package.
- Institution sandbox and implementation center.

## **Phase 4: Contextual expansion**

- Campus Hub.
- Housing/dining/billing status/deep links.
- Study Abroad.
- Athlete Plan.
- Portfolio/credentials.
- Advanced workspace integrations.

## Phase 5: Enterprise maturity

- Deep SIS/integration gateway.
- Multi-campus support.
- Advanced audit/permissions.
- Formal compliance program.
- Digital credentials and lifelong-learning record.

## 24. Exact AI coding-agent workflow

### Required first response from Claude Code / ChatGPT

Do not modify production code immediately. First:

1. Inspect the repository.
2. Create docs/AUDIT.md.
3. Create docs/SPRINT-1-REGISTRATION-PATH.md.
4. Create docs/DATA-MIGRATION-PLAN.md.
5. Create docs/SECURITY-GAP-ANALYSIS.md.
6. Create docs/ROUTE-AND-FEATURE-CROSSWALK.md.
7. Report exact files that would be changed before implementation.
8. Wait for confirmation before destructive refactors or production deployment.

## Branching

```
main                stable production
semester-unified-platform  integration branch
feature/action-center
feature/auth-supabase
feature/today-dashboard
feature/path-registration
feature/marketing-site
feature/study-studio
feature/institution-pilot
```

## Vertical slice workflow

```
Create feature branch
↓
Build one complete vertical slice
↓
Add persistence, permission checks, tests, ally states, loading/error states
↓
Deploy preview only
```

```
↓
Manual QA on mobile/desktop/keyboard/slow network
↓
Pull request and review
↓
Merge to integration branch
↓
Production release only with explicit approval
```

## Required test coverage

- Authentication and protected route behavior.
- Tenant isolation and object-level authorization.
- Action lifecycle and source precedence.
- Data source/freshness behavior.
- Schedule conflict calculations.
- Search-to-action flows.
- Accessibility: keyboard, focus, labels, dialogs, color contrast.
- Mobile/responsive behavior.
- External API/webhook failure behavior.
- Webhook idempotency/out-of-order event behavior.
- No secrets exposed in client bundle.
- Regression coverage for existing routes.

## 25. Exact first implementation prompt

Copy this into Claude Code after it audits the repository:

```
You are extending the existing Semester repository at:
https://github.com/harrisonjrubin7-cmyk/semester
```

```
This is not a rewrite. Preserve working routes, existing visual language,
existing tests, and existing state behavior unless a migration is planned,
tested, documented, and reversible.
```

```
First read:
```

```
- app/package.json
- app/README.md
- app/.env.example
- app/vercel.json
- app/vite.config.ts
- app/src/App.tsx
- app/src/main.tsx
- app/src/screens.tsx
- app/src/state/
- app/src/data/
```

- app/src/ai/
- app/src/intelligence/
- app/src/a11y/
- supabase/
- SEMESTER\_IMPLEMENTATION\_STATUS.md
- SEMESTER\_PRODUCT\_COMPLETENESS\_MATRIX.md
- SECURITY.md
- PILOT.md

Create these documents before changing product behavior:

- docs/AUDIT.md
- docs/SPRINT-1-REGISTRATION-PATH.md
- docs/DATA-MIGRATION-PLAN.md
- docs/SECURITY-GAP-ANALYSIS.md
- docs/ROUTE-AND-FEATURE-CROSSWALK.md

Then create branch feature/registration-path-pilot.

Implement only this first vertical slice:

- Supabase authentication and protected application routes if not already complete
- persistent profile and student profile data
- canonical Action model and action history
- Today dashboard with Path Snapshot, Next Best Step, and Immediate Commitments
- basic student-entered path / degree-plan persistence
- basic term planning and schedule conflict detection
- registration readiness checklist
- advisor meeting agenda with a safe view-only share/export option
- feedback capture and support route
- source labels: Institution verified, Imported, Student entered, Estimated, Needs review
- mobile responsive and accessibility-complete states
- analytics events for activation, plan creation, action completion, and clarity survey

Do not implement in this sprint:

- official registration execution
- deep SIS/LMS integrations
- payment, housing, dining, health, emergency, or parent portals
- full Study Studio
- unrestricted AI agents
- broad institutional analytics
- social/community feed

Use feature flags. Keep all AI and provider keys server-side. Do not expose secrets.

For every database table, add RLS policies. Students may only access their own records.

## 26. Definition of done

Semester is ready for an initial real pilot when:

- A student can securely create an account.
- A student can enter basic planning context.
- A student can see a transparent Path Snapshot.

- A student can build a term plan and identify a conflict.
- A student can save a backup course/option.
- A student can create an advisor meeting agenda.
- A student sees one clear next action and can understand why.
- A student can submit feedback or request help.
- The product works on mobile, keyboard, and screen reader basics.
- Data is persistent, tenant-aware, access-controlled, exportable/deletable.
- Demo versus verified data is clearly labeled.
- The team can measure activation, meaningful action completion, student clarity, advisor feedback, and support friction.
- The deployment has monitoring, backups, rollback plan, error logging, and support contact.

## **Final principle**

The strongest version of Semester is not the one with the most tabs or integrations. It is the one that consistently turns confusing academic and campus information into **one clear, explainable, safe next step**.